explicitly mention the importance and centrality of the costs and fees of annuities within DC plans. According to the DOL, those who are tasked with the responsibility of selecting annuities as a form of accumulation or distribution within a DC plan "…*must* consider the costs of an annuity contract in relation to the benefits and administrative services provided."

As far as the original study was concerned, the embedded guarantees promised at death were quite similar to equity put options, which are contracts that grant the holder a right but not an obligation to sell an underlying portfolio at a fixed price on a given date. These options are traded on financial exchanges all over the world and have readily available market prices. In the VA case, instead of the option maturing at a fixed date such as three weeks, months, or years in the future, the maturity date of the option was random; namely a date of death. Partially tongue in cheek, we called these securities, Titanic options.

Alas, our main research conclusion was that, if the extra added insurance fee was only meant to cover true risk, the typical VA policyholder was being grossly overcharged for the protection and peace of mind. We found that the basic RoP guarantee (that is, you get your money back at death) was worth no more than 5 to 10 basis points (or 0.05% to 0.10%) of assets per annum. Now, this obviously depended on the exact age of the policyholder, because older investors are more likely to die and hence cash in on the guarantee. Table 9.1 displays the theoretical model values of the death benefit guarantees for policy holders age 30 through 65.

TABLE 9.1 Capital Market Cost of a GMDB: RoP Guarantee

Age	Female	Male
30	0.3 b.p.	0.4 b.p.
40	0.8 b.p.	1.3 b.p.
50	2.0 b.p.	3.5 b.p.
60	5.0 b.p.	8.7 b.p.
65	7.6 b.p.	13.0 b.p.

b.p. = basis point

Assumptions: Equity: expected returns = 6%, volatility = 20%

Source: Moshe Milevsky and the IFID Centre, 2008.